JoyDeck 2 Hub

DIY DESKTOP CONTROLLER

DGVeLab

User Manual

Configuration and Operation

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1 Introduction, Safety and System Overview

1 1 JoyDeck 2 Description

JoyDeck 2 is a programmable desktop controller consisting of the JoyDeck 2 hardware and the JoyDeck 2 Hub Windows application. The physical joystick provides eight independent directions, each of which can be assigned a different command. The optional Page 2 function adds a temporary second set of eight commands while retaining the selected activation method. LED colors and overlay icons make assigned functions easy to identify.

1 2 Safety and Responsibility

JoyDeck 2 Hub can simulate keyboard and mouse input, launch programs, open links, send HTTP requests and execute system commands. Some configurations can therefore have immediate effects on the PC.

- Check a new configuration with Test action before using it routinely.
- Use particular care with shutdown, restart, logout and keyboard macros.
- Import configurations only from trusted sources.
- Save any open work before testing potentially critical commands.

1 3 Hardware

- Two-axis analog joystick, interpreted as eight directions.
- Integrated joystick button for confirmation and special button presses.
- Addressable RGB LED ring for visual feedback.
- ESP32 for input reading, LED management and USB serial communication.

1 4 Software

JoyDeck 2 Hub receives data from the device, applies calibration and the dead zone, interprets the direction, executes the configured action and manages persistent modes. This keeps the device simple while allowing its logic to evolve through software updates.

2 Interface Overview

The main window contains the connection controls, direction selector, command editor and configuration management buttons.

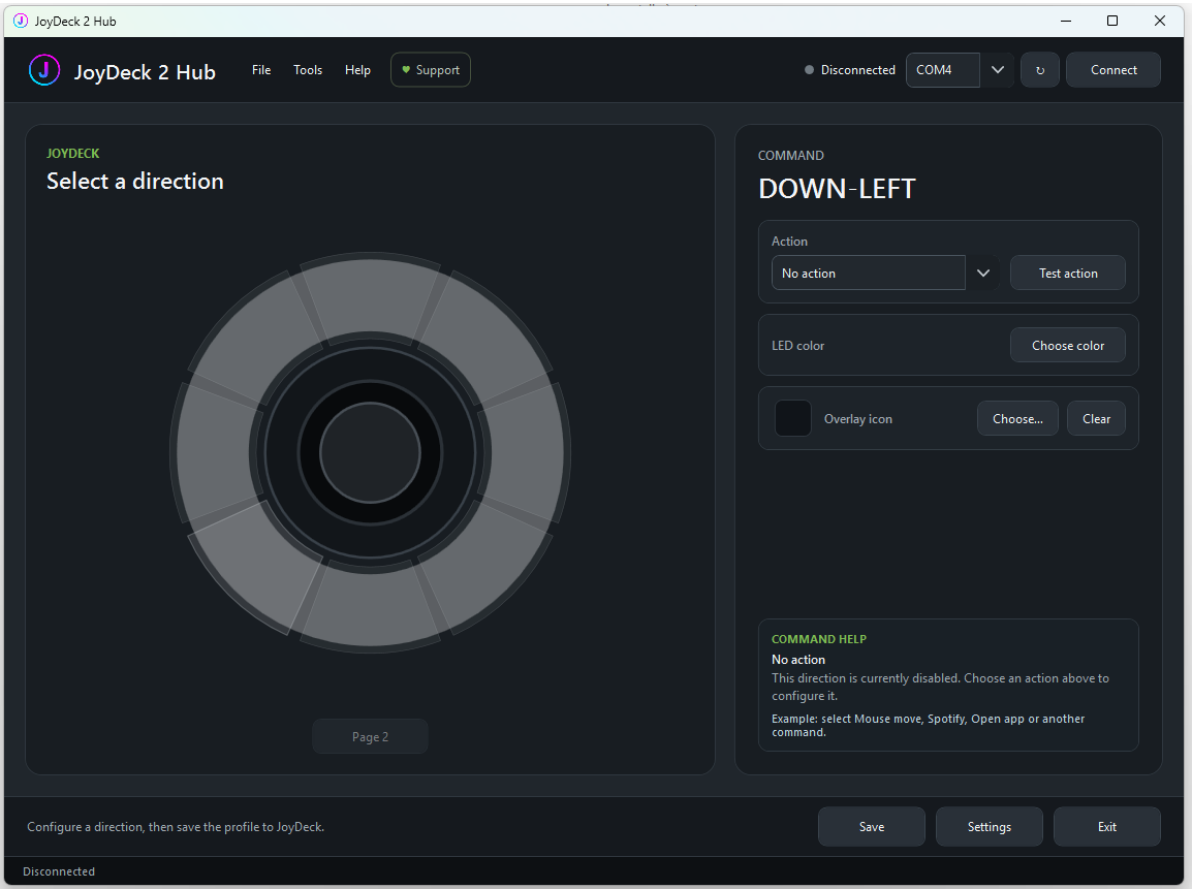


Figure 1 — JoyDeck 2 Hub main window.

2 1 Main Window Areas

Area	Available Controls
Top bar	File, Tools and Help menus, Support button and serial connection controls.
JOYDECK wheel	Eight sectors for selecting directions and the Page 2 button for the secondary page.
COMMAND panel	Action, command parameters, Test action, LED color and Overlay icon.
COMMAND HELP panel	Description, instructions and an example for the selected action.
Bottom bar	Save, Settings and Exit buttons, operational messages and connection status.

2 Connection and Direction Selection

2 2 Serial Connection Controls

The device connection controls are located in the upper-right corner of the window.

Control	Function
COM menu	Lists the available serial ports. Select the port associated with JoyDeck 2 before clicking Connect.
Refresh ports	The circular-arrow button refreshes the COM list. Use it after connecting the device if the port does not appear in the menu.
Connect	Opens the connection to the selected port, detects the device and performs calibration. Keep the stick centered during this process.
Disconnect	Closes the connection to JoyDeck 2. Replaces Connect when the device is connected.
Status indicator	The gray indicator with Disconnected indicates no connection. The green indicator identifies the connected state.

2 3 JOYDECK Selection Wheel

Click one of the eight wheel sectors to select the direction to configure. The selected sector is highlighted, and its name appears in the COMMAND panel heading, for example UP or DOWN-LEFT.

Selecting a sector on the Home screen opens the parameters for that direction. Use Test action to test the command. The associated color and icon identify the configured assignments.

2 4 Selecting Page 2

The Page 2 button below the wheel becomes available after the Page 2 action has been assigned to a direction on the main page. Click the button to view and configure the eight sectors on the second page.

Check the page and direction name before changing Action or its parameters. Page 1 and Page 2 use the same COMMAND editor; assignments for both pages are stored in the profile. The full procedure is described in Chapter 6.

2 Command Editor and Profile Management

2 5 COMMAND Panel

Control	Function
Action	Selects the command type to assign to the direction. No action disables the selected direction.
Action parameters	The fields change according to Action: application path, URL, key combination, mouse action, game type or BambuLab settings.
Test action	Tests the action with the configured parameters. Use it to check the result before saving.
Choose color	Opens the picker for the LED color associated with the direction. The color is also used to represent the sector.
Choose…	Opens the icon library in the Overlay icon section. Custom image… allows a personal image to be selected.
Clear	Removes the icon assigned to the direction without disabling the action.

2 6 Context Help and Status Messages

COMMAND HELP updates when an action is selected. It describes the command, provides usage instructions and gives a configuration example.

The lower message area shows operation results and configuration warnings. Check it after making changes, testing or saving. Warnings include assignment conflicts between Mouse move and Orbital. The status bar along the bottom edge displays serial connection information.

2 7 Menus and Management Buttons

File Contains Import configuration… and Export configuration… for loading and saving configurations and backup packages.

Tools Contains Restore factory settings to restore the initial configuration.

Help Provides commands to view the manual, check for updates, open the releases page and view program information.

Support Opens the project support page.

Save Saves the current configuration, including Page 1 and Page 2 assignments. Click Save after changing actions, parameters, colors or icons; Test action does not save the configuration.

Settings Opens the global settings: dead zone, mouse sensitivity, trigger, brightness, standby and startup with Windows.

Exit Requests that the application close.

3 Flashing the JD2 Firmware

3 1 Opening the Flash Center

1. Close JoyDeck 2 Hub and any applications using the device's serial port.

2. Connect JoyDeck 2 to the PC using a USB data cable.

3. Open Google Chrome or Microsoft Edge and access DGVeLab Flash Center using the following link.

DGVeLab Flash Center

4. Click DGVeLab Projects.

5. Open the Project menu and select JoyDeck 2.

6. Open the Firmware version menu and select the firmware version to install.

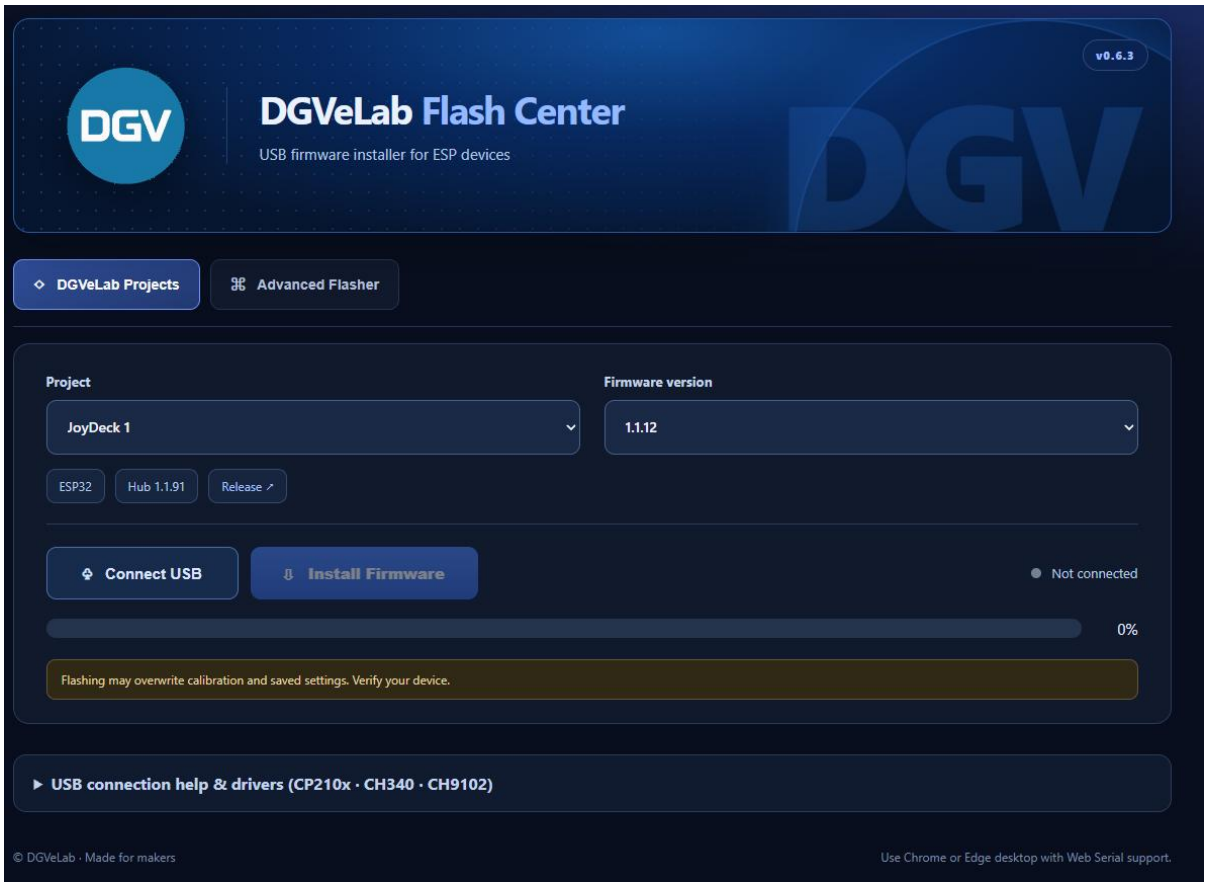


Figure F1 — Selecting the project and firmware version in DGVeLab Flash Center.

3 Writing the JD2 Firmware

3 2 Flashing the Device

1. Click Connect USB.

2. In the browser dialog, select the JoyDeck 2 serial port and click Connect.

3. Wait for the ESP32 to be detected and the Install Firmware button to become available.

4. Click Install Firmware.

5. In the confirmation dialog, check the JoyDeck 2 project and the selected firmware version, then click OK.

6. Wait for the progress bar to reach 100%, keeping the USB cable connected.

7. Wait for the Firmware written message to appear.

8. Close the Flash Center tab.

9. Disconnect the USB cable from JoyDeck 2 and reconnect it to the PC.

10. Start JoyDeck 2 Hub and complete the connection procedure in Section 3 7.

Connecting with the BOOT Button

1. Press and hold BOOT on the ESP32 board.

2. Click Connect USB and select the device port.

3. If the board is not detected, keep BOOT pressed and briefly press EN/RST.

4. Release BOOT once the ESP32 has been detected and continue with Install Firmware.

3 Downloading the USB Drivers

3 3 Selecting the Driver

1. In Flash Center, click USB connection help & drivers (CP210x · CH340 · CH9102).

2. Identify the USB-to-serial converter fitted to the ESP32 board.

3. Click the link for the converter: Silicon Labs — CP210x VCP, WCH — CH340 / CH341 or WCH — CH9102 / CH343 family.

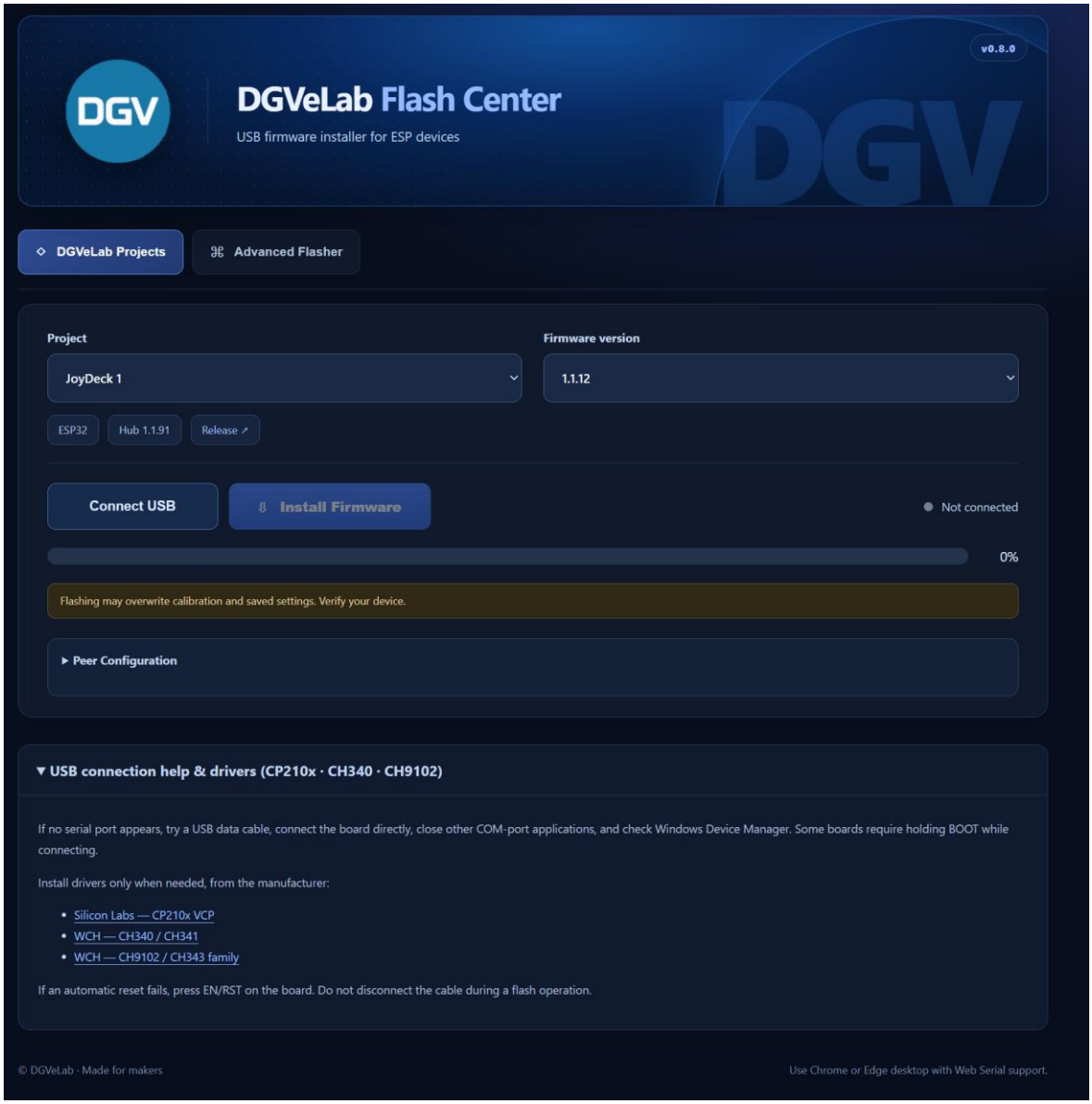


Figure F2 — USB serial driver links in the USB connection help & drivers panel.

3 Installing the USB Drivers

3 4 Downloading and Installing the Windows Package

1. Open the manufacturer's page using the link selected in Flash Center.

2. For CP210x, select Downloads and download CP210x Universal Windows Driver. For CH340/CH341, click Download on the CH341SER.ZIP page. For CH9102/CH343, click Download on the CH343SER.ZIP page.

3. Open the Windows Downloads folder, right-click the downloaded ZIP file and select Extract All.

4. Select the destination folder and click Extract.

5. With JoyDeck 2 connected via USB, right-click Start and select Device Manager.

6. Expand Other devices or Ports (COM & LPT) and select the JoyDeck 2 USB serial device.

7. Right-click the device and select Update driver.

8. Select Browse my computer for drivers.

9. Click Browse, select the folder extracted in Step 4 and confirm with OK.

10. Select Include subfolders and click Next.

11. Wait for installation to complete and click Close. Restart Windows if prompted.

12. Disconnect and reconnect the USB cable. In Device Manager, expand Ports (COM & LPT) and note the assigned COM port.

13. Return to Flash Center, click Connect USB and select the COM port identified in the previous step.

Manufacturer Download Pages

Silicon Labs CP210x VCP

WCH CH340 and CH341

WCH CH9102 and CH343

3 Flashing from a Firmware File

3 5 Loading JD2 Firmware with Advanced Flasher

1. Open the JoyDeck 2 Releases page using the following link and download the .dgv firmware file.

JoyDeck 2 Releases

2. Open DGVeLab Flash Center and click Advanced Flasher.

3. Click Connect USB, select the JoyDeck 2 port and confirm with Connect.

4. Wait for the ESP32 to be detected.

5. In the Firmware images section, click the File selector in the first row and open the downloaded .dgv firmware file.

6. In the Offset field in the same row, enter 0x0.

7. Open Flash settings and set Mode, Frequency and Size to Keep.

8. Click Flash Firmware.

9. In the summary, check the firmware file and address 0x0, then confirm with OK.

10. Wait for 100% and the Firmware write completed message, keeping the USB cable connected.

11. Close the Flash Center tab, then disconnect and reconnect the USB cable.

12. Start JoyDeck 2 Hub and complete the connection procedure in Section 3 7.

3 Starting JoyDeck 2 Hub

3 6 Installing and Starting the Program

- Use Windows 10 or Windows 11, 64-bit x64.
- Download and install the JoyDeck 2 Hub Setup package from the official release. The Windows program and ESP32 firmware are separate files.
- Complete the free activation process if required, and check that compatible firmware has been flashed.
- Close the flasher and any other application using the COM port before clicking Connect in Hub.

3 7 Step-by-Step Connection

1. Connect JoyDeck 2 to the PC.
2. Start JoyDeck 2 Hub.
3. Open the COM selector and choose the port associated with the device.
4. If the port is not listed, click the refresh button.
5. Click Connect.
6. Check that the indicator turns green and the device port is selected in the COM menu.

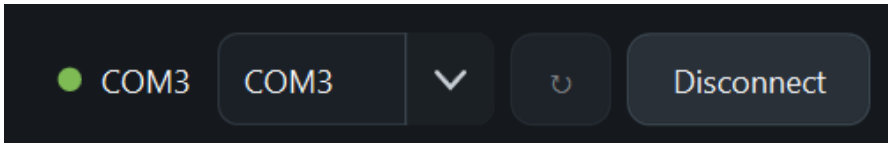


Figure 2 — Connected state: green indicator, active COM port and Disconnect button.

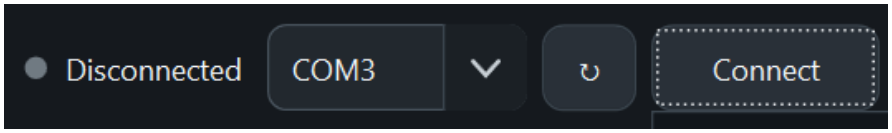


Figure 3 — Disconnected state: COM selection, port refresh and Connect.

3 8 Calibration

When a connection is established, JoyDeck 2 Hub calibrates the joystick center. Leave the stick completely untouched and centered during this process. Calibration compensates for joystick tolerances and works with the Dead zone setting to prevent unintended selections.

4 Configuring a Direction — Step-by-Step Guide

This procedure shows the complete workflow using Open app as an example. The same principle applies to all other actions: select the direction, choose the command, enter parameters, set the color and overlay, test and save.

4 1 Selecting the Direction and Command

1. Click the required sector of the wheel.

2. In the COMMAND panel, open Action.

3. Choose the command type. The relevant fields appear automatically.

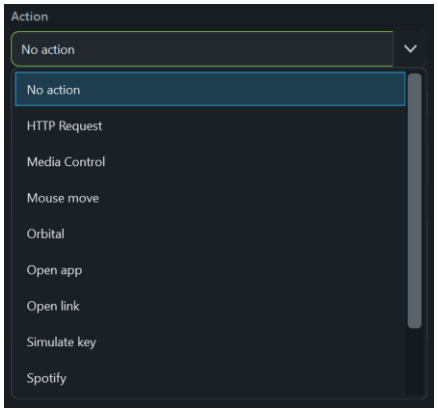


Figure 4 — List of available actions.

4 2 Entering the Parameters

Each action displays only the fields it requires. For Open app, for example, Application / file contains the path to the application or file; Paste pastes a path from the Clipboard and Browse... opens the file selection dialog.

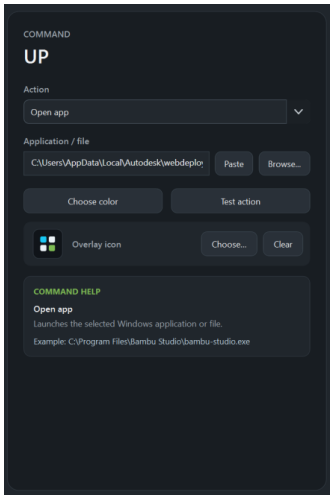


Figure 5 — Example of a direction configured with Open app.

4 Configuring the LED Color

4 3 Choosing the LED Color

Choose color opens the color picker. The selected color identifies the direction on the Home screen and is used for LED feedback. Use a consistent personal convention, for example green for applications, blue for mouse/CAD and cyan for multimedia.

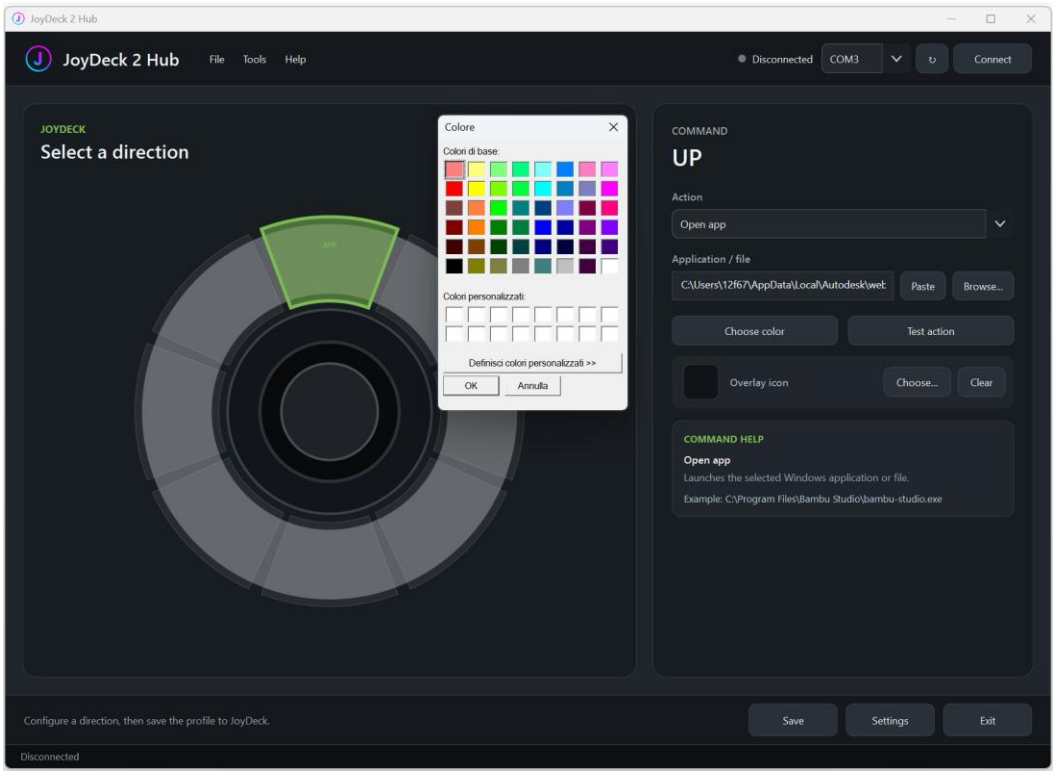


Figure 6 — Choosing the color for an Open app direction.

4 Configuring Icons and Saving

4 4 Assigning an Overlay Icon

Choose... opens the built-in icon library; Clear removes the icon. The library includes Media, Music, Browser, App, Mouse, Keyboard, System, Settings, Search, Download, Lock, Power, Wi-Fi, Game, Photo and User categories. Custom image... allows a personal image to be used.

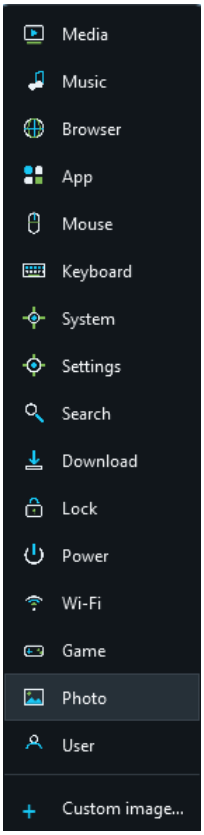


Figure 7 — Overlay icon library.

4 5 Test action and Save

Test action checks the command before it becomes part of the profile. If the result is correct, click Save. The COMMAND HELP panel remains available as a quick reference for the selected function.

5 No action

Disables the selected direction. Use it to leave a sector intentionally inactive without deleting or changing the other directions.

Practical Examples

Example 1 — Minimal Profile

Setup: assign the required commands to UP, DOWN, LEFT and RIGHT; set UP-RIGHT, DOWN-RIGHT, DOWN-LEFT and UP-LEFT to No action.
Result: the diagonals remain inactive, so an imprecise movement does not launch a command.

Example 2 — Reserved Direction

Setup: set DOWN-RIGHT, for example, to No action and leave the other seven commands unchanged.
Result: that position remains reserved for a future function without changing the rest of the profile.

6 Page 2

Page 2 extends the available commands without changing normal JoyDeck operation. Only one direction on the main page can be configured with Action = Page 2. When activated, JoyDeck temporarily switches to the second set of eight directions.

6 1 Configuration

1. Select an unassigned direction on Page 1.
2. Set Action = Page 2. The Page 2 action can be assigned to only one direction.
3. The Page 2 button below the wheel becomes available. Clicking it displays the second wheel on the Home screen and allows its eight sectors to be configured.
4. Page 2 uses the same global Trigger mode as Page 1 and has no independent triggers.
5. Page 2 provides the same actions as the main page, except for Page 2 itself.

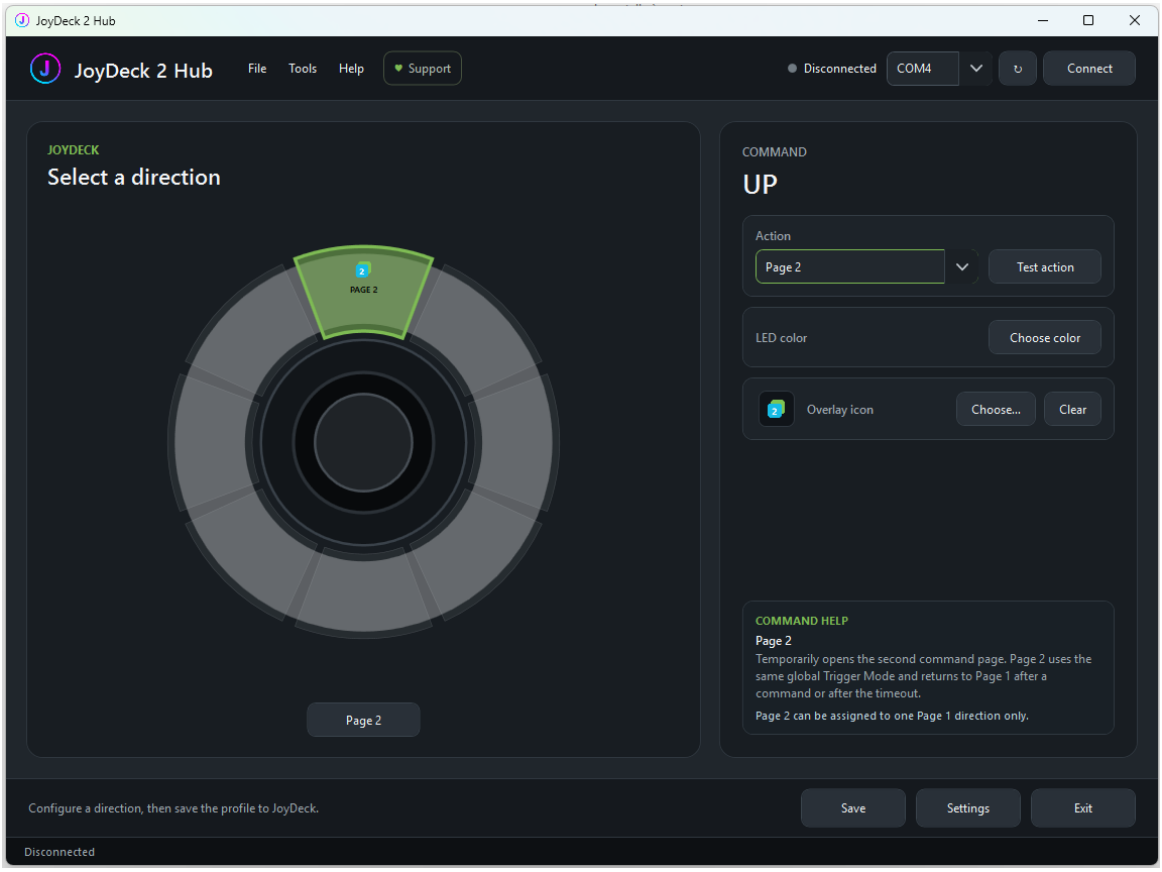


Figure 8 — Page 2 assigned to a direction on Page 1; the Page 2 button below the wheel becomes available.

Mouse move and Orbital are mutually exclusive across Page 1 and Page 2. Only one instance of either mode can be assigned in total; if one is already present, the other cannot be added.

The status information message explains Mouse move and Orbital assignment conflicts. Choose an unassigned direction or first remove the mode already configured.

6 Using Page 2

6 2 Operation

When the direction configured as Page 2 is activated, the second-page wheel appears in the lower-right corner. The joystick then selects one of the eight Page 2 actions using the same Trigger mode configured in Settings. After a command is executed, JoyDeck automatically returns to Page 1. If no command is selected within the timeout period, Page 2 closes and the main page is restored.



Figure 9 — Page 2 overlay during use.

Test action: when Action = Page 2, the Test action button displays the second-page overlay even if its eight sectors have not yet been configured. This allows its visual behavior to be checked immediately.

6 3 Saving

The Page 2 configuration is saved together with Page 1. ZIP backups also include the personal icons used on the second page; built-in icons are stored by reference. An older profile without Page 2 remains compatible and loads with an empty second page.

7 Open app

Launches a Windows application or opens a file using the configured path.

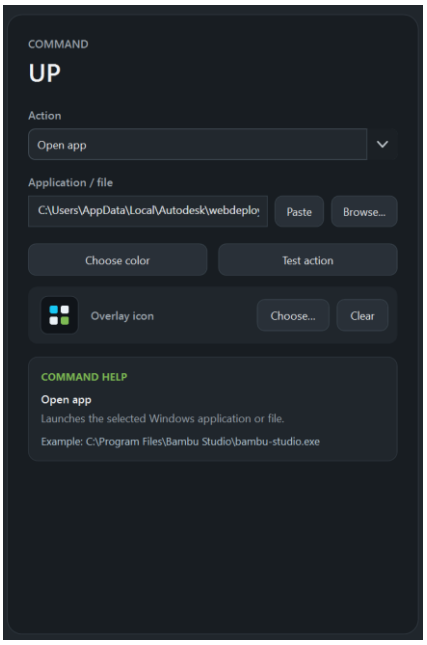


Figure 10 — Open app configuration.

Practical Examples

Example 1 — Bambu Studio

Setup: Application / file = C:\Program Files\Bambu Studio\bambu-studio.exe. If the path is different, use Browse... and select the executable installed on the PC.
Result: activating the direction launches Bambu Studio.

Example 2 — Work File

Setup: use Browse... and select a file, for example C:\Progetti\modello.3mf.
Result: activating the direction opens the file in the Windows application associated with the .3mf format.

8 Open link

Opens a web address in the default browser. The link should be complete, normally including the https:// prefix.

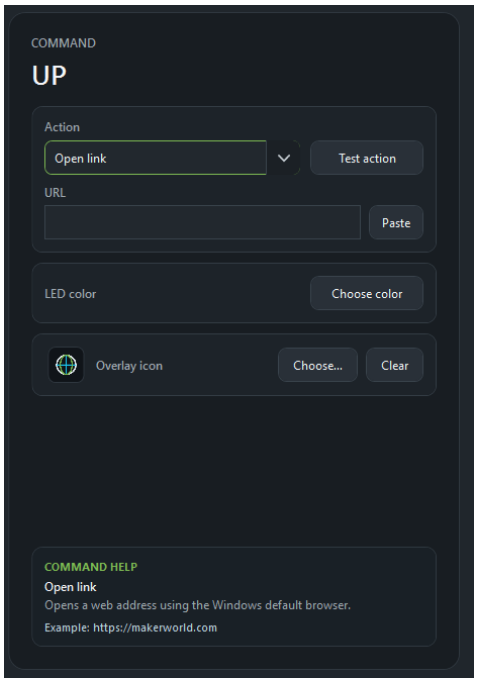


Figure 11 — Open link with the URL field and Paste button.

Practical Examples

Example 1 — MakerWorld

Setup: URL = https://makerworld.com/.
Result: activating the direction opens the page in the default Windows browser.

Example 2 — Local Control Panel

Setup: URL = http://192.168.1.50/ (replace the address with the device's actual address).
Result: activating the direction opens the device's web control panel on the local network.

9 System Control

Executes a system command provided by JoyDeck 2 Hub. Use it for Windows operations that need to be available without a keyboard.

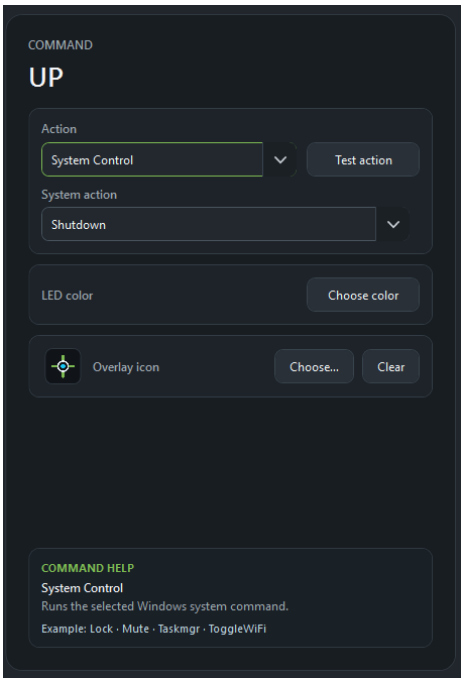


Figure 12 — System Control with the System action selector.

In the System action field, choose the Windows command to execute. Shutdown is selected in the screenshot: use Test action only after saving any open work.

Practical Examples

Example 1 — Volume

Setup: for UP, select System Control > VolUp; for DOWN, select System Control > VolDown.
Result: UP increases the Windows volume and DOWN decreases it.

Example 2 — Locking the PC

Setup: Action = System Control; command = Lock.
Result: activating the direction immediately locks the current Windows session.

Example 3 — Shutdown

Setup: Action = System Control; command = Shutdown. In Settings, choose On confirm or Hold as the Trigger mode.
Result: shutdown requires confirmation or a sustained input, reducing the risk of accidental activation.

Save any open work before testing shutdown, restart or logout.

10 HTTP Request

Sends an HTTP request to a web service, webhook or network device. This function integrates JoyDeck with external automations.

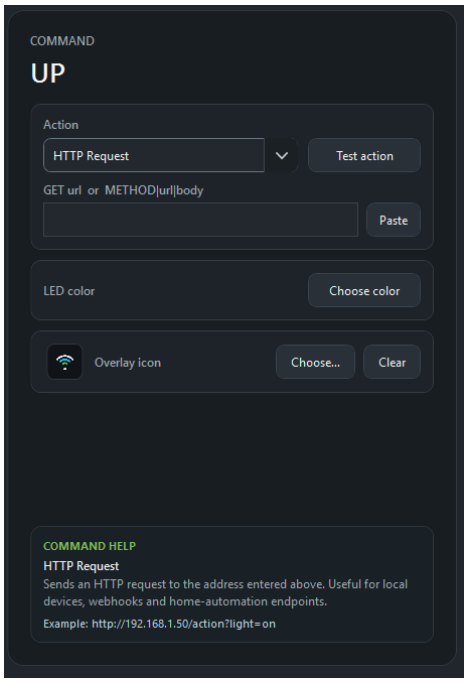


Figure 13 — HTTP Request with the GET url or METHOD|url|body field.

Practical Examples

Example 1 — GET

Setup: for a simple GET request, enter http://192.168.1.50/action?light=on in the GET url or METHOD|url|body field.
Result: JoyDeck sends a GET request to the specified address.

Example 2 — Webhook

Setup: for a POST request with JSON, enter POST|https://example.com/webhook/joydeck|{"event":"up"}. Replace the URL and content with those provided by the service.
Result: JoyDeck sends a POST request with a JSON body to the configured webhook.

Example 3 — ESP32

Setup: for an ESP32, enter an address such as http://192.168.1.80/relay/1/on.
Result: activating the direction calls the endpoint, and the microcontroller executes the associated function.

Do not share configurations containing tokens, credentials or private URLs.

11 Media Control

Activates a persistent mode for media controls. In this mode, the four cardinal directions control volume and tracks; diagonals are ignored.

Parameters and Behavior

- UP: volume up.
- DOWN: volume down.
- RIGHT: next track.
- LEFT: previous track.
- Long button press: exits the mode.

Practical Examples

Example 1 — Music While Working

Setup: Action = Media Control; leave Optional media URL / app empty if the player is already open.
Result: activating the direction enters controller mode: UP/DOWN adjust the volume, LEFT/RIGHT change tracks, a click toggles Play/Pause and a long press exits.

Example 2 — PC Connected to a TV

Setup: Action = Media Control; Optional media URL / app = spotify: to open Spotify automatically, if the protocol is registered in the system.
Result: JoyDeck opens Spotify and immediately switches to generic media controls; a long press closes the mode.

12 Simulate key

Simulate key sends keystrokes or key combinations. It provides three modes: Preset, Hold and Spam. Preset executes a shortcut once; Hold and Spam are persistent modes.

12 1 Preset

Shortcut contains ready-made key combinations. Custom... is available at the bottom of the list to enter a combination that is not in the library.

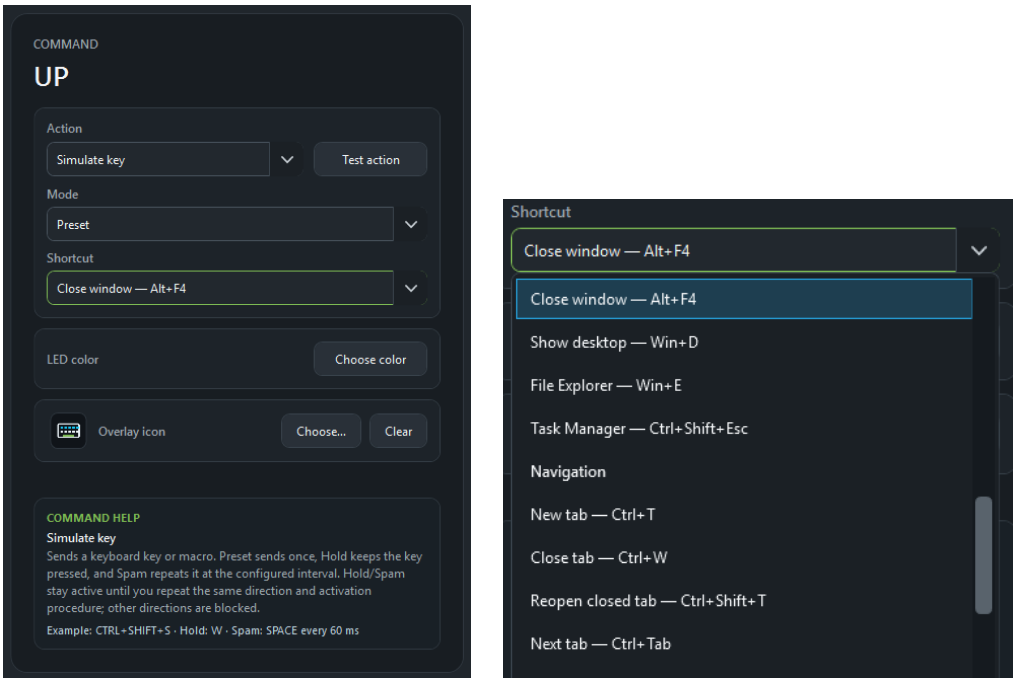


Figure 14 — Simulate key in Preset mode and a detail of the Shortcut list.

Practical Preset Examples

- Shortcut = Save as — Ctrl+Shift+S. Result: invokes the Save As command in the active application, if supported.
- Shortcut = Next tab — Ctrl+Tab. Result: switches to the next tab in the active browser.
- Shortcut = Custom...; Custom key / macro = CTRL+L; WAIT 200; TEXT https://makerworld.com; ENTER. Result: selects the address bar in the active browser, enters the URL and opens MakerWorld.

12 Simulate key — Hold and Spam

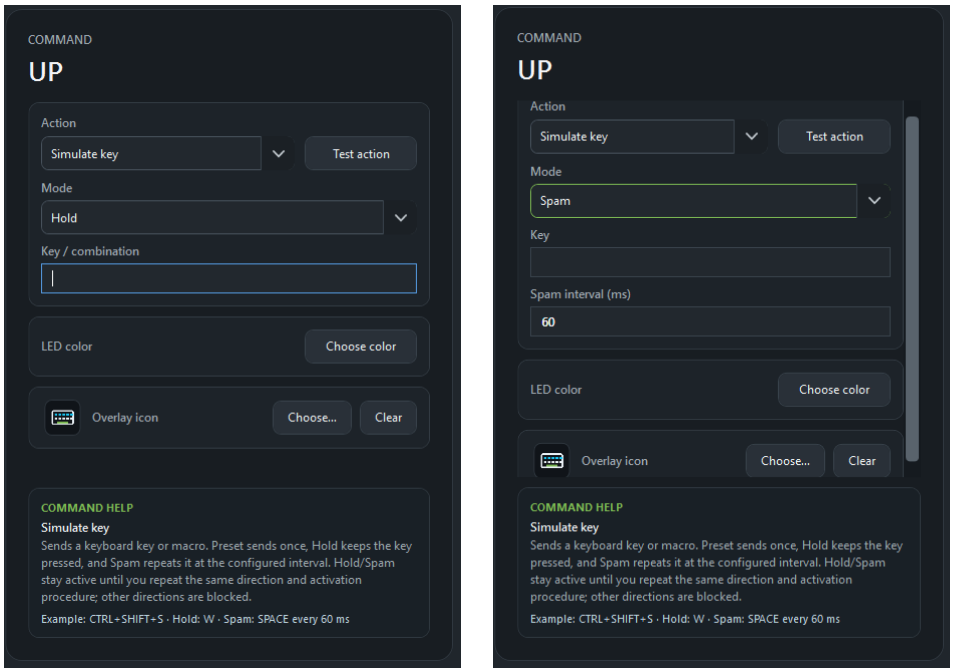


Figure 15 — Hold with Key / combination on the left, and Spam with Key and Spam interval on the right.

12 2 Hold

Keeps the key or key combination pressed. Once activated, it remains active even when the joystick returns to the center. To deactivate it, repeat the same direction and activation procedure.

Practical Hold Example

Setup: Mode = Hold; Key / combination = W.
Result: after activation, W remains pressed even when the joystick returns to the center. Repeat the same direction and activation procedure to release it.

12 3 Spam

Repeats the key or key combination at the configured interval. Spam also remains active until the same direction and activation procedure are repeated.

Practical Spam Example

Setup: Mode = Spam; Key = SPACE; Spam interval = 60 ms.
Result: JoyDeck sends SPACE every 60 ms until the same direction and activation procedure are repeated.

While Hold or Spam is active, the other directions are blocked. Spam interval applies only to Spam; Preset and Hold do not use it.

13 Mouse move

Temporarily turns JoyDeck into a pointing device. The joystick moves the cursor; speed depends on the distance from the center and the Mouse sensitivity value.

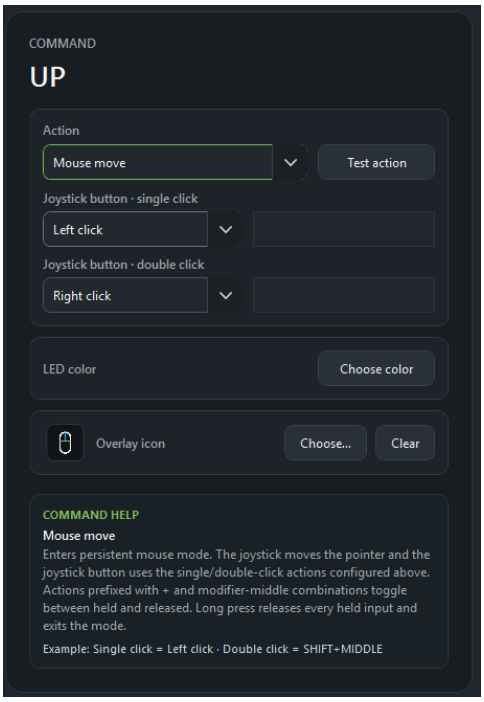


Figure 16 — Mouse move with separate single-click and double-click actions.

Joystick button · single click and Joystick button · double click define the button actions. Actions prefixed with + and combinations of a modifier with the middle button toggle between pressed and released states. A long press releases held inputs and closes the mode.

Parameters and Behavior

- Movement is proportional to the joystick position.
- Mouse sensitivity is adjusted in Settings.
- A long button press exits the mode.

Practical Examples

Example 1 — PC Control from the Sofa

Setup: Action = Mouse move; leave Single click = Left click and Double click = Right click. In Settings, set Mouse sensitivity = 10 as a starting value.
Result: the joystick moves the pointer; one joystick button click performs a left click and a double click performs a right click. A long press exits the mode.

Example 2 — Quick Adjustment

Setup: if the pointer is too fast, reduce Mouse sensitivity, for example from 10 to 7.
Result: for the same joystick movement, the cursor travels a shorter distance, making precise corrections easier.

14 Orbital

Orbital is designed for CAD, graphics and 3D environments. It moves the pointer while automatically holding down the selected mouse button.

Parameters and Behavior

- Orbital button: +Left click, +Right click or +Middle click.
- Delay: release delay in milliseconds.
- Long joystick button press: exits the mode.

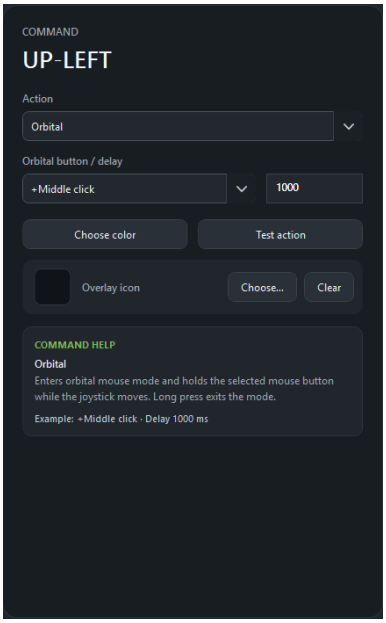


Figure 17 — Orbital configuration.

Practical Examples

Example 1 — CAD

Setup: Orbital button = +Middle click; Delay = 1000 ms.
Result: on entering Orbital, the joystick moves the pointer while holding down the middle mouse button; the configured delay is applied on release. A long press exits the mode.

Example 2 — Pan/Rotation

Setup: if the software uses the right mouse button for rotation, select Orbital button = +Right click; initially leave Delay = 1000 ms.
Result: JoyDeck performs the same movement while holding down the right button; adjust Delay if release is too fast or too slow.

15 Spotify

Activates a persistent control mode dedicated to Spotify. In the Spotify URI / URL field, paste a link copied directly from Spotify or use a compatible Spotify URI; JoyDeck opens the content and switches to playback control mode.

Parameters and Behavior

- Can open configured Spotify content.
- The mode remains active until it is exited.
- A long button press exits the mode.

Practical Examples

Example 1 — Playlist

Setup: in Spotify, open the required playlist and select Share > Copy link. Paste the link into the Spotify URI / URL field, for example https://open.spotify.com/playlist/... .
Result: activating the direction opens that playlist and enters Spotify mode; UP/DOWN adjust volume, LEFT/RIGHT change tracks or seek, a click toggles Play/Pause and a long press exits.

Example 2 — Music Controller

Setup: to open a single track, select Share > Copy link for the track and paste an address such as https://open.spotify.com/track/... into the Spotify URI / URL field.
Result: activating the direction opens the selected track in Spotify, and JoyDeck remains in controller mode until a long press is performed.

16 Mini Game

Mini Game turns JoyDeck into a persistent game mode managed by the PC, without changing the firmware. Select Action = Mini Game and choose Memory or Reaction in the Game type field. The LED color configured for the command identifies the mode when no game is in progress.

16 1 Starting and Exiting

Activate the direction assigned to Mini Game. The window displays the score and Hi-Score and prompts for a joystick button press. A click starts the first game or a new game after Game Over. The mode remains active between games; use a long button press to exit completely.

On entry and after each game, the LED ring returns to the color selected for the command. During play, the LEDs indicate directions: Memory uses cyan, Reaction uses amber, a correct response is confirmed in green and an error in red.

16 2 Memory

Memory displays a sequence of illuminated directions. Wait until the sequence ends, then repeat it with the joystick in the same order. Each completed round adds a new direction and increases the score. An incorrect direction ends the game.

16 3 Reaction

Reaction illuminates a direction and measures the time taken to reach it. The initial time limit is 1500 ms and decreases by 50 ms after each correct response, down to a minimum of 350 ms. The display shows the current limit, the last valid reaction time and the best recorded time.

If time runs out, the final result shows the limit and the measured time. If a different direction is selected, WRONG DIRECTION appears. After Game Over, the display alternates between the result and CLICK TO PLAY AGAIN.

16 Mini Game — Records and Position

16 4 Scores and Records

SCORE indicates the current game score; HI-SCORE stores the highest score for the selected game. Reaction also stores BEST REACTION, the shortest time achieved with a correct response.

Reset Hi-Score clears the selected game's records after confirmation. For Reaction, both the Hi-Score and best time are cleared. Records are included in ZIP backups and restored when the package is imported.

Practical Example

Setup: choose an unassigned direction, select Action = Mini Game, choose Reaction and assign a recognizable LED color. Click Test action, or save the profile and activate the direction. Click the joystick button to start; when the game ends, click again to replay or hold the button to exit.

16 5 Display Position

To position the Mini Game display on the desktop, drag it to the required location. The position is retained for subsequent openings.

17 BambuLab

BambuLab displays data on the PC from printers linked to a Bambu Lab account. The monitor shows progress, temperatures, fans and diagnostics in a dedicated window that can be opened from the joystick.

The connection uses Bambu Cloud. The PC must be on, JoyDeck 2 Hub must be running, JoyDeck must be connected via USB and an Internet connection is required. The printer must be available in the cloud account.

The function is for monitoring: it does not start or stop prints or change printer temperatures, speed or fans.

17 1 Signing In to Bambu Cloud

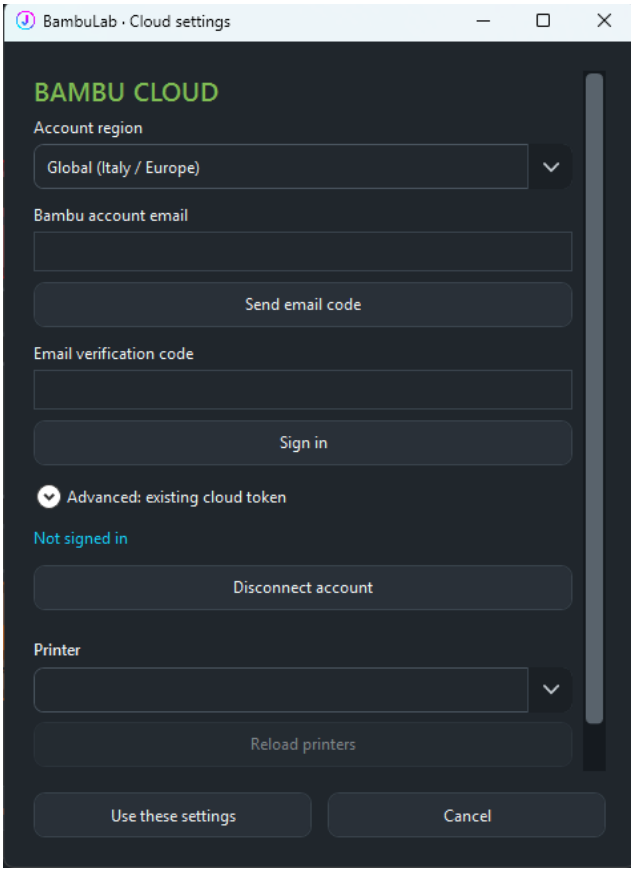


Figure 18 — Cloud settings before signing in to Bambu Cloud.

1. Select a direction on the Home screen, set Action = BambuLab and open Cloud settings.

2. In Account region, choose the account's region. For a European account, use Global (Italy / Europe), as shown in the example.

3. Enter the account address in Bambu account email and click Send email code.

4. Retrieve the code received by email, enter it in Email verification code and click Sign in.

17 BambuLab — Printer Selection

17 2 Assigning the Printer to the Direction

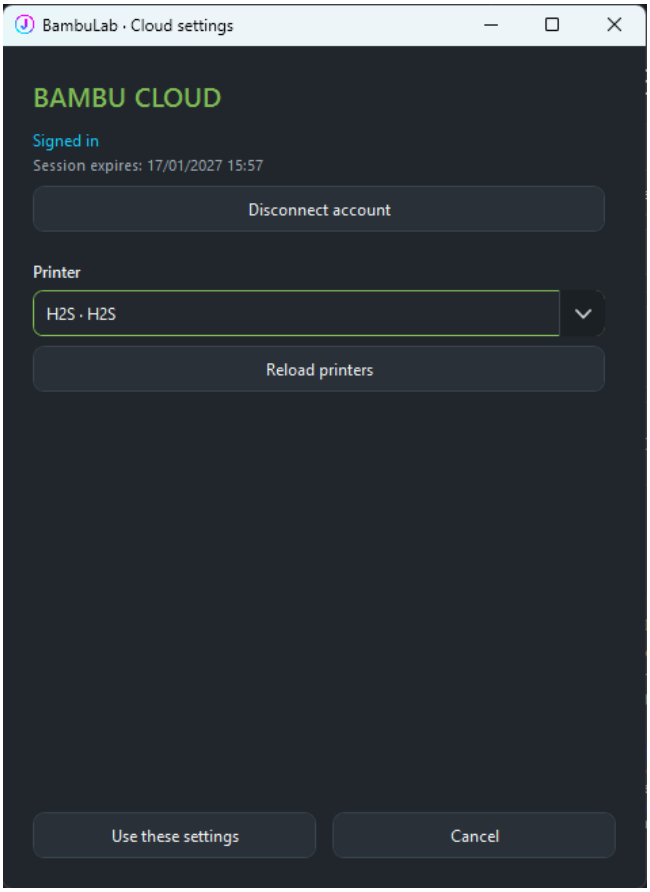


Figure 19 — Successful sign-in and printer selection in the Printer field.

1. Check that Signed in appears. Session expires shows the session expiry time, when available.

2. In the Printer field, choose the required printer. Reload printers refreshes the list of printers in the account.

3. Click Use these settings to apply the selection. Cancel closes the window and discards the selection.

4. On the Home screen, choose the LED color and overlay icon, then click Save. Use Test action to check that the monitor opens, and wait for the data.

The session is stored locally in protected form. If sign-in is no longer valid, repeat the procedure in Cloud settings. Disconnect account ends the account session.

Multiple Printers

BambuLab can be assigned to several directions, each with its own printer. Only one monitor is used at a time: to change printers, exit the current monitor and activate the other direction. The name in the heading identifies the printer being displayed.

17 BambuLab — Navigation and Progress

17 3 Controls During Use

Operation	Result
Activate the BambuLab direction	Opens the monitor for the assigned printer.
Click the joystick button	Switches to the next tab.
Long button press	Exits BambuLab mode.

The counter in the upper-right corner indicates the current tab and the total number available, for example 1 / 10. Available views depend on the data and functions exposed by the printer.

17 4 Print Progress

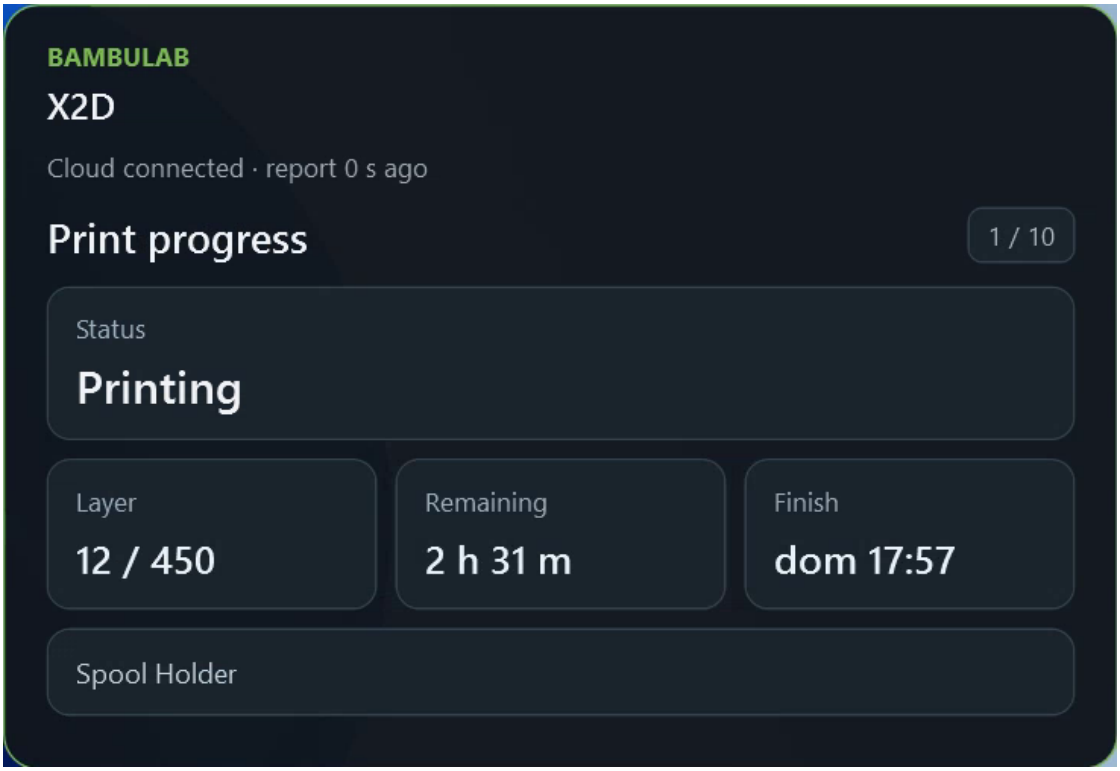


Figure B1 — Print progress during a print.

Print progress reports the printer status. Layer indicates the current layer and total layer count; Remaining shows the estimated time remaining; Finish indicates the expected completion time. When received, the job name appears at the bottom.

Times are estimates provided by the printer and may change during the job. In the heading, Cloud connected indicates the cloud connection; report … s ago shows how much time has elapsed since the last report was received.

17 BambuLab — Nozzle Temperatures

17 5 Reading Temperatures

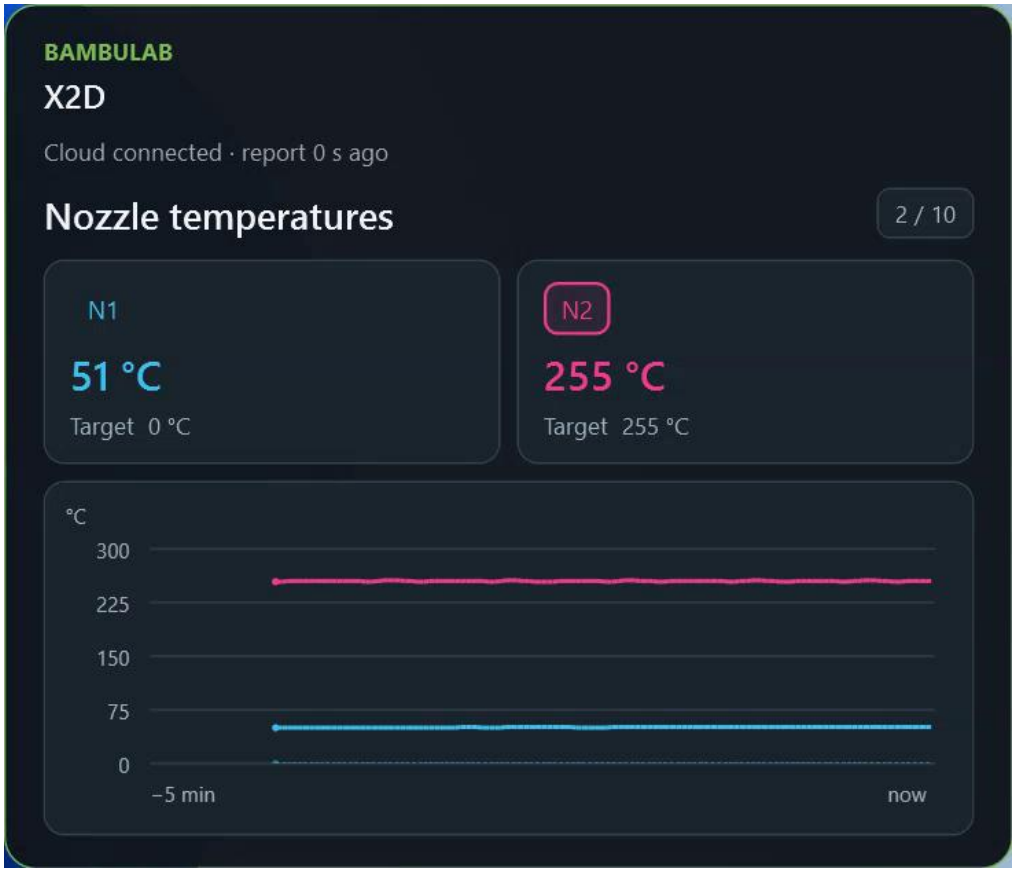


Figure B2 — Nozzle temperatures with two readings and their respective targets.

Nozzle temperatures shows the measured nozzle temperatures. Target is the temperature requested by the printer: a target of 0 °C may indicate that heating is not required for that nozzle, while the measured temperature may remain high during cooling.

On models with two nozzles, N1 and N2 distinguish the two readings. On the X2D, N1 identifies the left nozzle and N2 the right nozzle. The colors allow the graph curves to be followed separately.

Reading the Graph

The vertical axis shows degrees Celsius. The horizontal axis represents time and extends to the latest value, marked now. The window shown in the screenshots covers the last five minutes. The graph fills as data is received.

A zero target is different from an actual temperature of zero. Always compare the measured value with its corresponding Target when assessing a reading.

17 BambuLab — Bed and Chamber



Figure B3 — Bed temperature on the left and Chamber temperature on the right.

17 6 Bed Temperature

Bed temperature displays Current, the measured bed temperature, and Target, the value requested by the printer. The graph shows the heating phase and temperature stability during printing.

In the example, Current and Target are both 70 °C: the bed has reached the requested temperature. During heating or cooling, it is normal for the two values to differ.

17 7 Chamber Temperature

Chamber temperature displays the internal temperature reported by the printer and its trend over time. In the example, the chamber is at approximately 36 °C.

The availability of this measurement depends on the model and telemetry provided. Consult this tab when the printer exposes the sensor and its data.

Data Updating

When opened, the monitor may already have established the cloud connection without having received all the information. Wait for the full report before interpreting initial values, empty fields or missing tabs as actual printer readings.

17 BambuLab — Fans



Figure B4 — Part cooling fan on the left and Hotend fan on the right.

17 8 Fan Speeds

The fan tabs show Speed as a percentage and its trend over time. In the example screenshots, the part cooling fan runs at 33% and the hotend fan at 100%.

Tab	Monitored Fan
Part cooling fan	Part cooling.
Auxiliary fan	Auxiliary fan.
Chamber fan	Chamber fan.
Hotend fan	Hotend cooling.
Secondary auxiliary fan	Second auxiliary fan, when fitted.

Only information provided by the connected model is available. A speed of 0% may indicate a stopped fan; check it after the monitor has received updated data.

Changes in the graph reflect adjustments made by the printer during the job. The percentages are monitoring values, not controls for changing the speed.

17 BambuLab — Status and Display Position

17 9 Printer Status and Errors

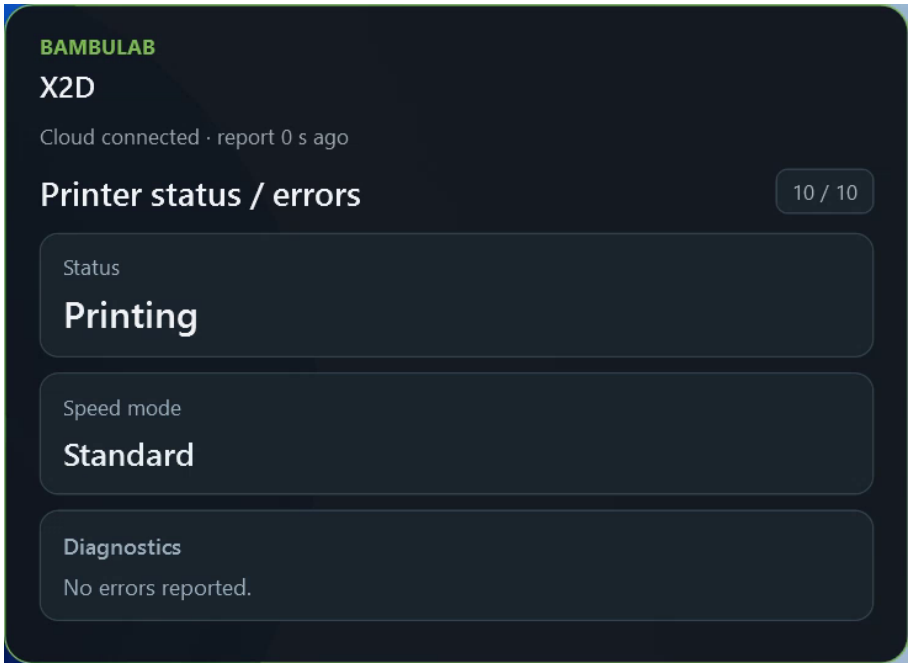


Figure B5 — Printer status / errors with status, speed mode and diagnostics.

Printer status / errors collects the operating status, the speed mode reported by the printer and diagnostic information. In the example, Status is Printing and Speed mode is Standard. No errors reported means that no errors are indicated in the report received.

If issues are reported, also check the printer display or Bambu Studio. The JoyDeck monitor provides a view of the data received from the cloud.

17 10 Display Position

To move the monitor, drag its heading to the required position on the desktop. The position is stored and restored the next time it opens, including after Hub restarts.

17 11 Connection and First Report

If the connection is active but some tabs are incomplete, leave the monitor open and wait for a full report. When reopening it shortly after closing, the wait may be up to approximately five minutes. Avoid repeatedly opening and closing it during this phase.

If updates still do not arrive, check the Internet connection, the printer's availability in the cloud and session validity in Cloud settings. Also check that the name in the heading matches the required printer.

18 Trigger mode

Trigger mode is a global setting: it determines how a selected direction becomes an executed command.

On release

The command executes when the joystick returns toward the center after selecting a direction. This is the factory setting shown in the current version.

On confirm

The direction is selected and then confirmed with the joystick button. Use this when explicit confirmation is required.

Hold

Requires the direction to be held for the specified time before activation. Use it for commands that should not start with a simple movement.

Practical Selection Examples

- On release — example: UP = Open app / Bambu Studio. Move the joystick to UP and return it to the center: the program starts on release.
- On confirm — example: RIGHT = System Control / Lock. Move the joystick to RIGHT and press the button to confirm: Windows locks.
- Hold — example: DOWN = System Control / Shutdown. Hold the direction until activation: a brief accidental movement does not start the command.

19 Settings

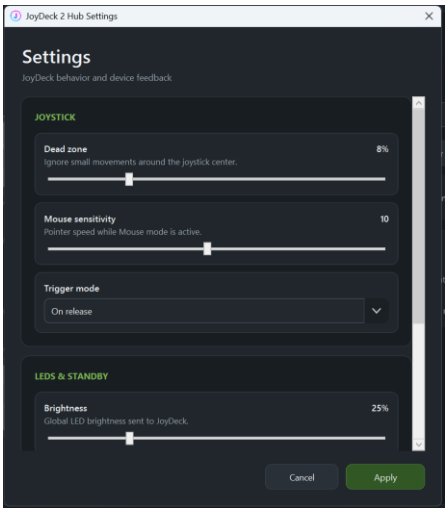


Figure 20 — Settings: Joystick, Trigger mode and Brightness.

Dead zone

Ignores small movements around the joystick center. Factory setting: 8%. Increase it if the joystick tends to select a direction while stationary.

Mouse sensitivity

Adjusts pointer speed in Mouse move and modes that use mouse movement. Factory setting: 10.

Trigger mode

Selects On release, On confirm or Hold for all directions.

Brightness

Adjusts the overall brightness of the LED ring. Factory setting: 25%.

Standby effect

Defines LED behavior when JoyDeck is idle. The factory setting is Disabled.

Start JoyDeck 2 Hub with Windows

Enables automatic application startup when signing in to Windows.

Practical Adjustment Example

Setup: start with Dead zone = 8% and Mouse sensitivity = 10. If a direction tends to activate on its own, try increasing Dead zone slightly; if Mouse move is too fast, try Mouse sensitivity = 7.
Result: joystick stability and pointer speed can be adjusted separately without changing assigned commands.

20 Configuration Management and Reset

Save

Saves the current profile. Use it after testing one or more changes.

Import and Export configuration

The File menu allows a configuration to be exported and imported later. The ZIP package is the complete backup: it includes Page 1, Page 2, custom icons from both pages, Mini Game Hi-Scores and the best Reaction time. Built-in icons are restored by reference. The JDF file contains the configuration without embedding personal icon files.

Restore factory settings

Tools > Restore factory settings returns the system to its initial values, clears Page 1 actions and empties Page 2.

- Actions: No action.
- Brightness: 25%.
- Dead zone: 8%.
- Mouse sensitivity: 10.
- Trigger mode: On release.
- Standby: Disabled.
- Auto Start: disabled.

Example

Setup: before creating a CAD profile, use File > Export configuration... and save a copy, for example JoyDeck_quotidiano.jdf. After testing, use File > Import configuration... to reload it.
Result: all directions and settings contained in the backup become available again without having to recreate them manually.

Before a reset, export the configuration to be retained. After an import, check Windows-related settings and reapply automatic startup if necessary.

21 Automatic Startup and Daily Use

JoyDeck 2 Hub can be configured to start with Windows. During daily use, the application can connect to the device and run in the system tray, keeping the controller available without keeping the main window open.

- At startup, leave the joystick centered during calibration.
- Check that the correct COM port is connected.
- Only one instance of JoyDeck 2 Hub should be running.
- To edit the profile, reopen the main window, make the changes and click Save.

Practical Routine Example

Setup: enable Start JoyDeck 2 Hub with Windows, save and leave JoyDeck connected via USB before shutting down the PC.
Result: at the next Windows sign-in, Hub can start, connect to the device and remain in the system tray; reopen the Home screen only when the profile needs to be edited.

22 Troubleshooting

JoyDeck Does Not Connect

Check the USB cable and COM port. Click the port refresh button; check that no other program is using the same serial port.

A Direction Activates on Its Own

Reconnect while leaving the joystick centered. Increase Dead zone if necessary.

The Command Does Not Execute

Check Action and its parameters, and click Save. Use Test action to distinguish a configuration problem from an activation problem.

Mouse, Orbital or Media Appears to Block Other Commands

A persistent mode is probably active. Use a long button press to exit.

Hold or Spam Remains Active

This is expected: repeat the same direction and activation procedure.

Open app Cannot Find the Program

Use Browse... and select the executable or file directly.

Automatic Startup Does Not Work After a Reset/Import

Check the option in Settings and apply/save it again so Windows receives the actual automatic startup setting.

22 Troubleshooting — Activation and BambuLab

Activation Error — Connection Timeout

First check that the firewall and antivirus allow JoyDeck 2 Hub to access the Internet. The problem may be temporary: close and restart the program, then repeat the activation request. If the error persists, check the Internet connection and try again later.

Page 2 Is Not Available

The Page 2 button on the Home screen remains disabled until a Page 1 direction is configured with Action = Page 2. Page 2 can be assigned to only one direction.

The BambuLab Monitor Does Not Open

Check that Action = BambuLab is assigned to the correct direction, that a printer has been selected and that the profile has been saved. Open Cloud settings to check sign-in, and try Test action.

The Cloud Is Connected, but Data or Tabs Are Missing

Wait for the full report while keeping the monitor open. An active cloud connection does not guarantee that all printer data has arrived. When reopening the monitor shortly after closing it, the wait may be up to approximately five minutes.

The Monitor Shows a Different Printer

Check the printer assigned to the direction being used. If several BambuLab directions are configured, check them individually and save. The name at the top of the window identifies the current printer.

23 FAQ

Can I Use JoyDeck Without Keeping the Main Window Open?

Yes. During normal use, Hub can run in the system tray, provided the application is running and the device is connected.

Can I Assign the Same Command to Several Directions?

Yes, subject to the restrictions of each action. Page 2 can be assigned to only one direction; Mouse move and Orbital are mutually exclusive and can be assigned only once in total. BambuLab can be assigned to several directions, each linked to the selected printer, with only one monitor active at a time.

What Is the Difference Between Trigger Hold and Simulate key Hold?

Trigger Hold is the method used to activate a direction. Simulate key Hold is a persistent action that keeps a key pressed.

How Do I Exit Hold or Spam?

Repeat the same direction and procedure used to activate the mode.

How Do I Exit Mouse move, Orbital, Media Control or Spotify?

Use a long press of the joystick button.

Can I Use Custom Icons?

Yes. In the Overlay icon selector, choose Custom image... .

How Do I Make a Backup?

Use File > Export configuration... and retain the exported file.

What Happens with Restore factory settings?

Actions are cleared and global settings return to the factory values listed in Section 20.

Can I Control IoT Devices?

Yes, if the device or service exposes an endpoint compatible with HTTP Request.

Why Do Diagonals Not Work in Media Control?

This is intentional: controller mode uses the four cardinal directions to make commands more distinct and predictable.

23 FAQ — Mini Game, Page 2 and BambuLab

How Do I Start a Mini Game and Exit the Mode?

Activate the configured direction and press the joystick button once to start or replay. Mini Game remains active after Game Over. Use a long press to exit completely.

What Is the Difference Between Memory and Reaction?

Memory requires watching and repeating an increasingly long sequence. Reaction requires reaching the illuminated direction quickly, within a time limit that decreases as the score increases.

Why Does Reaction Display WRONG DIRECTION?

The direction settled on by the joystick did not match the illuminated direction. The game applies a brief tolerance to ignore sectors crossed accidentally during movement.

How Do I Reset the Hi-Score and Best Time?

Select Mini Game and the required game, then click Reset Hi-Score and confirm. In Reaction, both the high score and BEST REACTION are cleared.

Are Mini Game Records Saved in the Backup?

Yes, when exporting the ZIP package. Importing that backup restores the Hi-Score and best Reaction time. An older package without records remains compatible and does not erase existing records.

How Does Page 2 Work?

Assign Action = Page 2 to a single direction on Page 1, then use the Page 2 button below the wheel to configure the eight secondary commands. During use, Page 2 remains active only until a command executes or the timeout expires.

Can I Use Mouse move or Orbital on Both Pages?

No. Mouse move and Orbital are mutually exclusive: only one instance of either mode can be configured across Page 1 and Page 2 combined.

Does BambuLab Work Without a PC?

No. The PC must be on, Hub must be running, JoyDeck must be connected via USB and a Bambu Cloud connection is required.

Can I Control the Print from JoyDeck?

The BambuLab function is for monitoring. It does not change heating, fans, movement or print status.

How Do I Change Tabs in the BambuLab Monitor?

Press the joystick button once. To exit the mode, hold it down.

Can I Assign Multiple Printers?

Yes. Configure a BambuLab direction for each required printer. Only one monitor is used at a time; check the printer name in the heading.